InfraCloud Project: Experimental Evaluation of Data Compression in Hadoop or Spark

1. Context and Objectives

Big Data frameworks such as Apache Hadoop and Apache Spark process large volumes of data distributed across multiple computing nodes. Their performance depends on several factors, including CPU capacity, disk I/O throughput, network bandwidth, and the amount of data exchanged between nodes.

Data compression is widely used to reduce storage requirements and communication overhead. However, compression and decompression introduce additional CPU costs that may negatively affect application performance. The benefits of compression therefore depend on the compression algorithm, the application workload, the characteristics of the infrastructure, and the volume of data being processed.

Compression can be applied at two different levels:

- **Storage-level compression:** Data is compressed before being stored in HDFS, reducing disk space requirements and potentially improving disk I/O performance.
- **Communication-level compression:** Intermediate data exchanged between distributed tasks, particularly during shuffle operations, is compressed to reduce network traffic.

The objective of this project is to **experimentally evaluate the performance impact and computational cost of different compression algorithms in Hadoop and Spark**, considering both storage and communication compression.

Students are expected to conduct a rigorous experimental study, produce performance curves, and identify the conditions under which compression improves or degrades application performance.

2. Experimental Infrastructure

Students will deploy and configure a distributed Hadoop and Spark environment on a cluster of virtual machines.

The infrastructure must include:

- A distributed HDFS storage system.
- Hadoop MapReduce or Apache Spark execution environments.
- Multiple worker nodes for distributed computation.
- Monitoring tools to measure CPU utilization, memory usage, disk I/O, and network traffic.

Students must ensure that the experimental environment is reproducible and that the same infrastructure configuration is used when comparing compression algorithms.

3. Applications and Workloads

The experimental evaluation must be conducted using **three distinct distributed applications**, each representing a different workload profile:

- **Application 1 — I/O-bound:** An application whose performance is primarily limited by disk I/O or data transfers, with relatively low computational requirements.
- **Application 2 — CPU-bound:** A computationally intensive application whose execution time is primarily dominated by CPU operations rather than data movement.
- **Application 3 — Balanced CPU/I/O:** An application that involves significant computation and data movement, without either resource being the dominant bottleneck.

Each application must be evaluated using either Hadoop MapReduce or Apache Spark, with equivalent processing logic to ensure meaningful comparisons.

Students must experimentally characterize the three applications and demonstrate, using CPU, disk I/O, and network measurements, that they exhibit the expected workload characteristics.

For each application, students must evaluate multiple compression algorithms, different dataset sizes, and both storage-level and communication-level compression.

The objective is to determine **how the computational characteristics of an application influence the performance benefits and overhead of compression**.

4. Compression Algorithms

Students must evaluate several compression algorithms supported by the selected Hadoop or Spark configurations.

The study should include, where supported:

- **Snappy:** Optimized for fast compression and decompression.
- **LZ4:** Designed for high compression and decompression throughput.
- **Gzip:** Typically provides a higher compression ratio at a greater computational cost.
- **Zstandard (Zstd):** Offers configurable trade-offs between compression ratio and computational overhead.

An execution without compression must be used as the baseline.

Students must first evaluate the intrinsic characteristics of the selected algorithms, including compression ratio, compression time, decompression time, and CPU cost.

5. Experimental Evaluation

The experimental study must investigate the following dimensions.

5.1 Storage-Level Compression

Students must evaluate the impact of compressing data stored in HDFS.

For each compression algorithm, they must measure:

- The size of the compressed dataset.
- The compression and decompression overhead.
- The execution time of the applications.
- CPU utilization and consumption.
- Disk I/O volume and throughput.

Experiments must be performed with **different dataset sizes** to evaluate how compression overhead and benefits evolve as data volume increases.

Students must explain whether reducing disk I/O compensates for the additional CPU cost introduced by compression.

5.2 Communication-Level Compression

Students must evaluate the impact of compression on intermediate data exchanged between distributed tasks, particularly during Hadoop MapReduce or Spark shuffle operations.

The evaluation must measure:

- Total application execution time.
- Volume of data transferred over the network.
- Network throughput.
- CPU overhead associated with compression and decompression.
- Shuffle execution time.

Students must determine whether reducing network traffic improves overall performance and identify situations in which compression becomes counterproductive.

5.3 Combined Compression

Students must also investigate the interaction between storage-level and communication-level compression.

At least four configurations must be compared:

Configuration	Storage compression	Communication compression
C0	Disabled	Disabled
C1	Enabled	Disabled
C2	Disabled	Enabled
C3	Enabled	Enabled

Students must determine whether enabling compression at both levels provides additional benefits or introduces unnecessary computational overhead.

5.4 Impact of Dataset Size

All major experiments must be repeated using different input dataset sizes.

The selected sizes must be sufficiently different to reveal changes in application behavior and compression efficiency. For example, datasets of 1 GB, 5 GB, 10 GB, and 20 GB may be considered, depending on the available infrastructure.

Students must investigate whether the relative performance of compression algorithms changes as the amount of data increases.

6. Expected Results and Performance Analysis

The main deliverable of this project is a **quantitative experimental evaluation supported by performance curves**.

Students must produce graphs illustrating at least the following relationships:

Performance curve	Comparison
Execution time vs. dataset size	Different compression algorithms
CPU consumption vs. dataset size	Compression overhead
Disk I/O volume vs. dataset size	Storage compression
Network traffic vs. dataset size	Communication compression
Compression ratio vs. algorithm	Compression efficiency
Execution time vs. compression configuration	C0, C1, C2, C3
Execution time vs. workload type	I/O-bound, CPU-bound, mixed

These experiments must be conducted for the selected applications and either Hadoop or Spark (you select), whenever the corresponding compression mechanisms are supported.

Each experiment must be repeated several times to account for performance variability. Graphs must include appropriate units, legends, and variability indicators.

Students must not limit their work to presenting experimental curves. They must analyze and explain the observed behavior.

In particular, their analysis should address the following questions:

1. Which compression algorithms provide the best trade-off between compression ratio and CPU overhead?
2. Does compression systematically improve the performance of I/O-bound applications?
3. How does compression affect CPU-bound applications?
4. How do the benefits of compression evolve with increasing dataset sizes?
5. Is compression more beneficial for storage operations or network communications?
6. Can combining storage and communication compression degrade performance?
7. Under which conditions should compression be disabled?

Students must support their conclusions with experimental evidence rather than general assumptions about the compression algorithms.

7. Deliverables and Evaluation

Each group must submit:

1. Experimental infrastructure: A functional Hadoop or Spark environment, including deployment scripts, configuration files, and monitoring tools.

2. Experimental results: Raw measurements, experiment scripts, and all performance graphs. The results must be reproducible.

3. Technical report: A report describing the experimental methodology, workloads, compression configurations, performance results, and analysis. Particular attention must be paid to explaining the observed compression overhead and the differences between workloads.

4. Oral presentation: A presentation of the main findings, supported by experimental curves. Students must be able to justify their experimental choices and explain unexpected results.